GEOMATICS NTNU

Deadline: 13.03.2023
Individual report. Compulsory.
Evaluation: Passed/not passed

TBA4245 GEODESY, Assignment no. 2:
Network reliability analysis

Relevant material (Blackboard):

- Lecture notes Network Analysis parts 1-4.
- “Multippel T-test for quality control” by Jon Glenn Gjevestad. On Blackboard, Course materials, Support literature folder

Report: Individual report. Submit the report and computer code on the Blackboard.

Tasks:

1. A gross error test in a levelling network.
2. Internal and external reliability analysis of the network.
3. Write computer code to perform the tasks.

See the solution calculated by GISLINE, printouts with estimated gross errors (“Est. grovfeil”) and reliability analysis in the three last pages.

The height difference between a known benchmark A with height 0.000 m and an unknown point B are levelled four times. For the second levelling, more accurate equipment was used. Observations (unit mm) and weights are shown in the ***l***-vector and the **P**-matrix:

$$\boldsymbol{l} = \begin{bmatrix} 100100 \\ 100099 \\ 100101 \\ 100111 \end{bmatrix} \qquad \mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

1) Calculate the weighted average $\hat{l} = \frac{\sum p_i l_i}{\sum p_i}$, the corrections $v_i = \hat{l} - l_i$, the matrices **v = Ax – f** , and $\sum p_i v_i^2 = \mathbf{v^T P v}$

2) From 1) we have $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}^T$ and **f =** ***l***. Find the Least Squares estimate $\hat{\mathbf{x}} = \mathbf{(A^T PA)^{-1} A^T Pf}$ (matrix form). Comment on your result.

3) Calculate "delete-one" estimates of gross errors in each observation with methods:
   1) $\nabla_i = l_i - \frac{\sum p' l'}{\sum p'}$ where *p'* and *l'* are all the *l* and *p*, except l_i and p_i.
   2) $\nabla_i = l_i - \hat{\mathbf{x}}_i$ where $\hat{\mathbf{x}}_i = \mathbf{(A^T P_i A)^{-1} A^T P_i f}$ are computed putting a very low weight on observation no. *i*. For example, $p_i = 0.000001$.
   3) ∇_i as in part 2: Blunder estimation in the note from *Jon Glenn Gjevestad* (NMBU)

   Compare the results and explain which of the three methods you will prefer.

4) Calculate the redundancy matrix of the observations as: $\mathbf{R = I - A(A^T PA)^{-1} A^T P}$

5) Calculate the co-factor matrix of the observation corrections from: $\mathbf{Q_{vv} = RP^{-1}}$

6) Calculate the 4 co-factors $q_{\nabla\nabla}$ of the 4 gross errors ∇_i

   The **R-**matrix is generally not diagonal. That means a specific blunder will affect all estimated corrections. It is common to simplify the case, extracting diagonal elements only.

Law of Error Propagation (LEP):

Given:

1) Mathematical linear relations between two random vectors **x** and **y: x** = **Dy** + **c**
2) the covariance matrix $\mathbf{C}_{yy}$ of **y**

Then the covariance matrix of **x** can be derived as: $\mathbf{C_{XX} = DC}_{yy}\mathbf{D^T}$

With the mathematical approximation $\nabla \approx \frac{-1}{r} v = \frac{-1}{p\, q_{vv}} v$ (See also Leick 2004 eq. (4.366), pp 157)

and then LEP, we find: $$q_{\nabla\nabla} = \left(\frac{-1}{q_{vv}\, p}\right) q_{vv} \left(\frac{-1}{q_{vv}\, p}\right) = \frac{1}{q_{vv}\, p^2}$$

Instead of following the single element calculations in the solution, it is from now on recommended to use the matrix-based method described in the note "*Multippel T-test*" by *Jon Glenn Gjevestad.* You will get identical numerical results as in the solution.

7) Estimate the standard deviations of unit weight in the 4 cases

8) Estimate the standard deviations for the 4 gross errors.

9) Calculate the 4 test statistics values: $t_i = \dfrac{\hat{\nabla}_i}{\hat{\sigma}_{\nabla_i}}$

10) Compare the test values with your chosen Student-T table value.
    Choose a significance level of $\alpha = 5\%$ for the total test.
    Calculate the individual significance level: $\alpha_i = 1 - (1 - \alpha_{TOT})^{1/n}$ for the test of 1 observation?

11) Make your conclusions: Did you find any gross error?

Internal reliability on the 4 observations:

12) Interpret the redundancy matrix you calculated in 4.

13) Estimate the largest remaining gross error.

Tip: Use the confidence interval for the largest remaining gross error for the 4 observations:

$$\left[\hat{\boldsymbol{\nabla}} - \hat{\boldsymbol{\sigma}}_{\boldsymbol{\nabla}}.\boldsymbol{T}_{\alpha/2,f}\ ,\ \hat{\boldsymbol{\nabla}} + \hat{\boldsymbol{\sigma}}_{\boldsymbol{\nabla}}.\boldsymbol{T}_{\alpha/2,f}\right]$$

External reliability analysis on the 4 observations:

It is recommended to use the matrix-based method described in the note “Multipple T-test for quality control” by Jon Glenn Gjevestad.

14) Calculate and explain the external reliability in this 1D case (levelling) for the 4 gross errors.

GISLINE documentation:

```
NYBESTEMTE KOORDINATER MED MIDLERE FEIL [meter]. *Estimated coordinates*

PUNKT                X           Y           H         sX       sY       sH
--------------------------------------------------------------------------
B                                        100.1020                  0.0026

KORRIGERTE OBSERVASJONER, ANTATT MIDLERE FEIL OG RESTFEIL [meter / gon]
*Corrected observations. Assumed standard deviation and residuals*

Fra           Til            Observasjon    Antatt M   Restfeil
From          To             Observation    Assumed SD Residual
-----------------------------------------------------------------------
A            1 B         dH      100.1000     0.0010     0.0020
A            2 B         dH      100.0990     0.0007     0.0030   *
A            3 B         dH      100.1010     0.0010     0.0010
A            4 B         dH      100.1110     0.0010    -0.0090   ***

STATISTIKK

Antall iterasjoner                         :            1

Antall observasjoner høydeforskjell        :            4
Antall observasjoner                       :            4   No. of obs.
Antall ukjente høydekoordinater            :            1
Antall ukjente                             :            1   No. of unknowns

Antall overbestemmelser                    :            3  Degree of freedom

Feilkvadratsum                             :          104.00542964   sum PVV
Beregnet std.avvik på vektsenheten                 5.8880    Estimated SD
Antatt   std.avvik på vektsenheten         :            1.0000 Assumed SD of
                                                  the obs. with weight p = 1


TEST AV OBSERVASJONER - MULTIPPEL T-TEST

Fra           Til               Restfeil  Est.grovfeil   Testverdi
From          To                Residual  Est. gross error Test value
------------------------------------------------------------------
A            1 B         dH       0.0020     -0.0025         0.32
A            2 B         dH       0.0030     -0.0050         0.90
A            3 B         dH       0.0010     -0.0012         0.16
A            4 B         dH      -0.0090      0.0113         8.58

Tabellverdi=8.77 (Student-t, f=2, alfa=0.0064)
Table value
```

```
OPPSUMERING ETTER TEST AV OBSERVASJONER: *Summary, gross error test*

Kategori                 Ant.obs.   Akkumulert (%)
------------------------------------------------------------------------
      Test/Tabell < 1.0          4             100.00  No gross error
1.0 < Test/Tabell < 2.0          0             100.00
2.0 < Test/Tabell < 3.0          0             100.00
3.0 < Test/Tabell < 3.0          0             100.00
Ukontrollerbar                   0             100.00

STATISTIKK

Antall iterasjoner                       :            1

Antall observasjoner høydeforskjell      :            4
Antall observasjoner                     :            4
Antall ukjente høydekoordinater          :            2
Antall tilleggsukjente                   :            2
Antall ukjente                           :            4

Rangdefekt                               :            3
Antall ukjente korrigert                 :            1

Antall overbestemmelser                  :            3

Feilkvadratsum                           :          104.00542964
Beregnet std.avvik på vektsenheten               5.8880
Antatt   std.avvik på vektsenheten       :            1.0000


TEST AV M0

Tabellverdi    =      7.82 (Kjikvadrat, f=3, alfa=0.0500)
Beregnet verdi =    104.01 ***
```

Ingen feil i observasjonsmatrialet er funnet

NO GROSS ERRORS FOUND

Printout from GISLINE, reliability of the unknown height of B:

The Norw. land surveying program Gisline gives the same result as in the calculations above. From the Gisline report file (dok file):

```
YTRE PÅLITELIGHET - KOORDINATER [meter]
External reliability – Coordinates

KOORDINAT Observasjon [meter/gon]    Indre pål.       Ytre pål.
                                     Internal reliab. External
reliab.
---------------------------------------------------------------------
--
H B        A        2 B         dH    -0.0289          -0.0116
```

Explanation/interpretation
of the GISLINE report above:

It is observation no. 2 from A to B with a largest remaining gross error (LRE) of -0,0289 m that will give the largest **<u>change/deformation on the final height of B with -11.6 mm = -0,0116 m.</u>**